



Retardant Is Not Just for Airtankers Anymore

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Ask any wildland firefighter what retardant looks like and they will describe the same thing. A large airtanker low on final, a billowing red-orange curtain dropping across a ridgeline, the satisfying sound of artificial rain as thousands of gallons of fire retardant hit the fuel bed below. That image is burned into the collective memory of this profession. It is also, increasingly, a limiting one.

Long-term fire retardant has been an aerial tool for more than 60 years, and the chemistry behind it has steadily improved with each generation of product development. Today's formulations are rigorously tested through the USDA Forest Service Wildland Fire Chemical Systems program and listed on the agency's Qualified Products List, with active ingredients classified by the EPA as practically non-toxic for mammals (*USDA Forest Service, Wildland Fire Chemical Systems, Qualified Products List, 2024*). **What has not changed is where the profession applies this chemistry.** The hard reality is that we have been leaving one of our most effective suppression tools sitting on the shelf while our fires burn hotter, move faster, and grow larger, often to levels that make traditional direct attack untenable. Ground application of long-term retardant is not a new concept. It is an underused one. And closing that gap requires something more difficult

than ordering product. It requires a genuine shift in how the wildfire community thinks about retardant in the first place.

What the Chemistry Does

Long-term fire retardant works through a thermal chemical chain reaction that alters the combustion pathway of treated fuels. Unlike water, which suppresses fire by absorbing heat, or foam, which creates a temporary barrier against oxygen and penetrates the fuel and ground by breaking water surface tension, long-term retardant changes the molecular structure of the fuel itself. When the water carrier evaporates, the phosphate residue remains active in the fuel bed. When flame contacts treated vegetation, the phosphate promotes char formation

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rather than flaming combustion, generating water vapor and graphite-like carbon instead of the volatile gases that sustain a flame front. The retardant does not go away when the water dries. It stays put through wind, heat, and smoke, remaining effective for days, weeks, and months, until it is diluted by rainfall or washed away mechanically.

This persistence is the property that makes ground application so strategically valuable. A retardant line placed in the right location ahead of a fire greatly slows or stops spread depending on fuel type, and it can often hold without crew presence. That is a fundamentally different proposition than a wet line, a hose lay, or a hand line in marginal terrain, all of which require people to stand between the fire and the objective. Ground-applied retardant can be placed on critical infrastructure, around structures and subdivisions, and values at risk with pinpoint accuracy, hours or days ahead of a wildfire.

The Perception Problem

The obstacle to ground application is not technical. It is cultural. In most academies, fire training, and formal classes, retardant is covered in the context of aviation. There is no meaningful training on the full spectrum of wildland fire chemicals, which includes water, Class A foam, gel, and long-term retardant. Firefighters at all levels learn how to calculate airtanker drop zones, how to communicate with air attack, and how to avoid being under a retardant drop. Ground application is rarely, if ever, mentioned as an equivalent tactical option. The result is a generation of wildfire professionals, and more critically a generation of company officers, fire line supervisors, and operations section chiefs, who think of retardant as something ordered from the air branch and delivered by a pilot. The idea of loading the same product into a water tender or a retardant delivery trailer and applying it from a hose line or hose lay does not fit their mental model of fire suppression.

That mental model is reinforced by operational culture. Airtankers are prestigious, highly visible, and politically popular. Ground retardant application looks like an engine company or hand crew doing what they usually do, just with a different product. There is no air show. There is no one stopping work to take pictures. The work is quiet and the results are not immediate, because long-term retardant is designed to work when the fire arrives, not while crews are watching.

This perception gap has real consequences. When the aerial window closes because of low visibility, wind, or darkness, suppression and protection options using retardant collapse for agencies that have not integrated ground application into their doctrine. The fire, often, does not stop running at 2000 hours. The engine company that knows how to pre-treat a subdivision during a night shift has options that a crew waiting on an airtanker at first light does not.

The Case for Timing

The most compelling argument for ground application is not that it replaces aerial retardant. It is that it works when aerial retardant cannot, and it works before the fire forces the decision.

Research published in the Proceedings of the National Academy of Sciences by Stanford researchers (*Appel et al., 2019*) examined what happens when retardant is applied proactively to high-risk fuel beds rather than reactively once a fire is running. The findings were striking. Proactive treatment of a given area required roughly 20,000 gallons of retardant, compared to an estimated one million gallons needed to achieve comparable suppression effect after a fire was actively spreading. The researchers were not making an argument against aerial suppression. They were demonstrating the power of timing. Getting the right product on the ground before a fire arrives is fundamentally more efficient than waiting for the fire to force the decision, and a ground-based delivery system gives incident managers the ability to act during the hours when aircraft cannot fly.

The platforms required are already on every ramp and in every station that runs a wildland engine program. Long-term retardant can be applied through standard water tenders, Type 3 and Type 6 engines, retardant application trailers, brush trucks, UTVs with tank kits, and backpack sprayers for structure perimeter work. There is no specialized equipment requirement. The only requirements are that the product is pre positioned and that the crew knows how to use it.

There is an additional operational reality worth naming directly. Aircraft availability on significant fire weather days is finite. When multiple fires are running simultaneously, air resources are triaged, contracts have daily flight hour limits, and large airtankers are committed well in advance of where the next ignition will occur. Ground application does not compete with that system. It complements it. A crew that has pre-treated a subdivision or a contingency line with long-term retardant the night before has already accomplished what would otherwise require multiple airtanker loads, freeing those aircraft to work the active fire perimeter where their speed and reach cannot be replicated from the ground. Ground retardant does not replace air. It multiplies what air can do by covering the ground it cannot.

Bobcat Fire, Sierra Madre, 2020

The most instructive recent example of ground application done right is not a large agency program or a federally funded pilot study. It is a municipal fire department in the San Gabriel Valley. Rich Snyder spent 36 years at the Sierra Madre Fire Department before retiring as the city's fire marshal. During that career he developed one of the most sophisticated ground retardant programs in the country, applying long-term retardant annually to protect high-risk interface zones throughout the San Gabriel Valley, including a standing pre-treatment program around Pasadena's Rose Bowl stadium before each Fourth of July fireworks event. That institutional knowledge was put to the test in the fall of 2020.

When the Bobcat Fire, which started in the Angeles National Forest, made a run toward the communities of Sierra Madre, Arcadia, and Monrovia, Snyder, then working as a consultant for Allied Disaster Defense, coordinated the deployment of PHOS-CHEK long-term fire retardant from ground-based apparatus ahead of the fire's arrival. The three cities ordered three totes

of 265 gallons each of concentrate. Working while aerial assets were limited by smoke and darkness, crews pre-treated structure perimeters and fuel transitions in critical interface zones. Among the protected targets was the historic Trask Boy Scout Camp in the foothills above Monrovia. The camp had been vacant during COVID and the vegetation around the cabins and infrastructure had not been maintained to its usual standard. By conventional assessment it was a likely loss. Sierra Madre Fire mixed approximately 1,800 gallons of retardant and applied it around the camp's buildings and infrastructure. Several days later the fire made it into the canyon. None of those structures were lost. The fire could not make it past the retardant lines.

Sierra Madre did not accomplish this because they had better equipment or more personnel. They accomplished it because their department had been integrating ground retardant application into their WUI doctrine for more than 30 years. The product was on the apparatus, the crew knew the mix ratios and application rates, and the company officer knew where to put it. **When the aerial window closed and every other agency was waiting on aircraft, Sierra Madre was still working.**

Despite this success, ground-applied retardant is still not fully integrated into initial attack planning. When incident commanders build their resource orders at the start of an incident, ground application should be part of that conversation, not an afterthought called in once aerial options are exhausted. Ground-based units can be pre-positioned ahead of a moving fire with priorities focused on protecting critical infrastructure, keeping evacuation corridors open, treating planned firing operations, and pre-identifying Safety Zones and Temporary Refuge Areas before conditions make that work impossible from the air.

The Window We Are Wasting

Prescribed fire is among the most resource-intensive operations the wildfire community undertakes. A single burn project can require months of planning, multiple environmental clearances, air quality authorizations, agency coordination, crew scheduling, and pre-burn preparation. Burn bosses and prescribed fire incident commanders walk the unit repeatedly. Holding resources are staged. Contingency plans are written for contingency plans. All of that preparation happens before a single ignition is made.

And yet, in nearly all of that preparation, ground application of long-term retardant along the burn perimeter is not part of the plan.

It should be. The same lead time that makes prescribed fire projects manageable is exactly the window that allows for thorough, deliberate retardant pre-treatment of the highest-risk perimeter segments. Structures near the unit boundary, roads and fences that serve as values at risk, interface areas where an escape would immediately threaten homes or infrastructure, all of these can be treated in the days before ignition with the same product that has been protecting values from wildfire for decades. The retardant does not interfere with the prescribed fire. It sits on the fuels outside the intended burn area and waits. If the fire stays in the unit, it was cheap insurance.

If the fire escapes, it may be the only thing standing between a successful burn and a federal disaster declaration.

The history makes the case. In 2000, the Cerro Grande Fire began as a prescribed burn on Bandelier National Monument in New Mexico. It escaped, burned into the community of Los Alamos, destroyed more than 200 homes, forced the evacuation of 18,000 residents, and ultimately cost the federal government more than 875 million dollars in suppression, damages, and claims settlement (*National Park Service, Cerro Grande Fire After-Action Report, 2001*). In the spring of 2022, two prescribed burns in northern New Mexico merged to become the Hermits Peak and Calf Canyon Fire, the largest fire in New Mexico history. It burned more than 341,000 acres, destroyed over 900 structures, and generated damages that prompted Congress to authorize more than 3.7 billion dollars in relief funding for affected residents. The Forest Service paused all prescribed fire operations nationally in the aftermath (*USDA Forest Service, Hermits Peak and Calf Canyon Fire Review, 2022*). The list of costly and devastating prescribed fire escapes is extensive.

The same failure to use an available ground window is not limited to prescribed fire. The Dragon Bravo Fire at the Grand Canyon's North Rim in July 2025 made that point in the most visible way possible. The fire ignited on July 4 from a lightning strike. Fire managers had nine days before the blaze ran into the developed area on the night of July 12, destroying the historic Grand Canyon Lodge, a National Historic Landmark that had stood since 1928, along with 106 surrounding structures, at a suppression cost of 124 million dollars. One firefighter, J. Hank Hester, lost his life during the incident. When the lodge was finally threatened, aerial retardant was blocked on two fronts: a chlorine gas leak at the water treatment facility made drops chemically dangerous, and interagency retardant avoidance areas under the Red Book prohibited drops in the lodge's vicinity. A discretionary property protection exception existed in NPS policy. It was never invoked. Ground-applied retardant, pre-positioned on the structure perimeters and fuel transitions around the lodge complex during the nine days between ignition and loss, was not part of the plan. **The aerial window was closed. The ground window had been open for over a week.** (*National Park Service, Dragon Bravo Fire Update, July 13, 2025; AZ Backroads Special Report, November 2025*)

These were not isolated failures of planning or forecasting. They were failures to fully exploit every available tool to hold the line when conditions pushed fire toward the objective. Ground-applied retardant, pre-positioned on the high-risk segments of those perimeters before the fire arrived, would not have guaranteed a different outcome in every case. But it would have changed the calculus at the moment the fire tested the line. The cost of pre-treating a burn perimeter or a threatened structure complex with retardant is measured in hundreds or low thousands of dollars. **The cost of the losses described above is measured in the billions.**

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How many more historic structures and lookout towers will we lose before we use the tools already available to save them?

Prescribed fire burn associations, VMP coordinators, and fuel officers who want to expand the use of beneficial fire, who are fighting every year against public skepticism, liability concerns, and the political fallout from high-profile escapes, have a tool available to them that reduces the consequence of a worst-case scenario. Using it is not an admission of doubt in the burn plan. It is the same logic that puts holding resources on every corner of the unit. You stage the engines because you are serious. You apply the retardant because you are thorough.

The Paradigm Shift That Is Required

Adopting ground application as a standard suppression tool is not a matter of buying product and writing a new section into the SOG. The cultural inertia is real and it runs deep. Wildfire professionals trust what they trained on. Incident commanders order what their operations section chiefs request. Operations section chiefs request what their branches and division supervisors know how to deploy. At every level of the system, **ground retardant application is a capability that exists theoretically but rarely gets ordered because nobody in the chain has made it habitual.**

Breaking that cycle requires explicit action at the training level. Ground application of long-term retardant should be included in engine academy curricula alongside foam operations. Division and group supervisor classes and operations section chief courses should be teaching the coordinated use of a ground-applied workforce, pre-treating around subdivisions and critical infrastructure, and reinforcing planned firing operations with ground-applied retardant lines. The profession should consider formalizing this capability through a dedicated functional group, a Ground Applied Retardant Group, with task forces working under it as a recognized ICS structure. It should be tested. It should be expected.

It also requires procurement decisions made before fire season, not during it. Pre positioning product at the station or warehouse level in high-risk WUI jurisdictions is the foundational step. **A department that does not have retardant on hand when the alarms sound for an initial attack fire cannot improvise it into existence.**

Incident management teams should be asking the question explicitly during long-term fire planning: are there structures, secondary lines, or critical infrastructure that we can pre-treat with retardant while aircraft are engaged on the main fire? That question should appear on size-up worksheets, in strategy and tactics meetings, the ICS-215 planning process, and in operational planning sessions. It should be part of the pre-shift briefing during elevated fire weather.

Contingency groups deserve particular attention in this conversation. On large incidents, especially Type 1 and Type 2 events, contingency groups are a standard ICS assignment. They work well out in front of the

main fire perimeter, constructing line, prepping critical infrastructure, and establishing the secondary positions that the incident will fall back to if primary lines are breached. That is exactly the operational environment where ground-applied retardant belongs. A contingency group pre-treating structure clusters, road corridors, and fuel transitions ahead of the fire's projected spread is doing something that no crew with hand tools or dozer alone can replicate. The retardant stays active when the crew moves to the next assignment. It works through the night when the group is off the line. And it is still holding when the fire tests the contingency position the following afternoon. Incorporating a retardant application function into contingency group operations on large incidents is not a significant leap in doctrine. It is a logical extension of work those resources are already doing, with a tool that makes that work last.

The 24-hour operational window that ground application provides, uninterrupted by inversion layers, aircraft maintenance cycles, darkness, or the resource competition that defines most significant initial attack situations, is a tactical advantage the wildfire community has largely chosen to ignore.

The Bottom Line

Retardant works. The aerial delivery system that has defined it for six decades works too, and it remains an irreplaceable part of the modern suppression toolkit. But the fire environment we are operating in now generates rates of spread, flame lengths, and spotting distances that regularly outpace reactive suppression. The full potential of retardant chemistry is not realized when it is limited to delivery from aircraft. It is realized when ground crews treat the right fuels at the right time, in the hours and locations where aircraft cannot operate, filling the gaps that every incident management team knows exist but rarely has a tool to address.

The wildfire professionals who understand this are already using it. The departments that have made ground application a standard capability are protecting structures and critical infrastructure that would otherwise be lost. Prescribed fire practitioners who integrate retardant into their perimeter plans are reducing the exposure that has eroded public trust more than once. The rest of the profession is one paradigm shift away from the same outcome.

The retardant has been there all along. The question is whether we are willing to apply it from the ground. 

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